

Host A/B suite — netkit vs TF Lite

Apple Silicon · warm-only · MNIST CNN + MNIST DS-CNN + MobileNetV4-Conv-Small (ImageNet) · models + fixture images excluded from flash/RAM

1. Runtime flash / RAM (absolute)

MCU-style image only: netkit ELF TEXT/DATA minus fixture .o; TF Lite = core LiteRT CPU libs. Same TF footprint for int8 and float32.

XNNPACK ON		TF÷nk ≈ 9.4× / 3.9×		XNNPACK OFF		TF÷nk ≈ 64× / 48×	
1.31–1.32 MiB	netkit flash	12.41 MiB	TF Lite flash	193–200 KiB	netkit flash	12.41 MiB	TF Lite flash
191.8 KiB	netkit RAM	752.0 KiB	TF Lite RAM	15.8 KiB	netkit RAM	752.0 KiB	TF Lite RAM

mode	netkit flash	TF flash	flash TF÷nk	netkit RAM	TF RAM	ram TF÷nk
XNNPACK ON	1.31–1.32 MiB	12.41 MiB	9.4×	191.8 KiB	752.0 KiB	3.9×
XNNPACK OFF	193–200 KiB	12.41 MiB	~64×	15.8 KiB	752.0 KiB	47.7×

Models + methodology

MNIST CNN = digit classifier. MNIST DS-CNN = depthwise-separable digit peer. MNv4-Small ImageNet = MobileNetV4-Conv-Small (10-class fixture). Discard 1st process; order swaps. MNIST: discard run 0 + image 0. ImageNet: discard full first image pass. Ratio = TF ÷ netkit (>1 ⇒ netkit faster / smaller).

2. INT8 — all models (absolute + TF÷netkit)

model	XNN	netkit	TF Lite	latency	flash	ram
MNIST CNN	ON	20.118 μs	20.463 μs	1.017×	9.4×	3.9×
MNIST DS-CNN	ON	20.560 μs	21.121 μs	1.027×	9.4×	3.9×
MNv4-Small ImageNet	ON	681 μs	713 μs	1.048×	9.4×	3.9×
MNIST CNN	OFF	153 μs	578 μs	3.78×	63.5×	47.7×
MNIST DS-CNN	OFF	196 μs	424 μs	2.17×	63.5×	47.7×
MNv4-Small ImageNet	OFF	7.62 ms	28.8 ms	3.78×	65.6×	47.7×

3. FLOAT32 — all models (absolute + TF÷netkit)

model	XNN	netkit	TF Lite	latency	flash	ram
MNIST CNN	ON	30.582 μs	31.532 μs	1.031×	9.4×	3.9×
MNIST DS-CNN	ON	26.860 μs	29.300 μs	1.091×	9.4×	3.9×
MNv4-Small ImageNet	ON	1.088 ms	1.171 ms	1.076×	9.4×	3.9×
MNIST CNN	OFF	508 μs	1.01 ms	1.99×	65.8×	47.7×
MNIST DS-CNN	OFF	272 μs	445 μs	1.63×	65.8×	47.7×
MNv4-Small ImageNet	OFF	32.7 ms	61.5 ms	1.88×	63.6×	47.7×

Green: netkit ahead on latency (ratio ≥ 1). Amber: TF Lite ahead or tied within noise (< 1). Source: host_ab_suite_results_int8.txt / host_ab_suite_results_float32.txt.